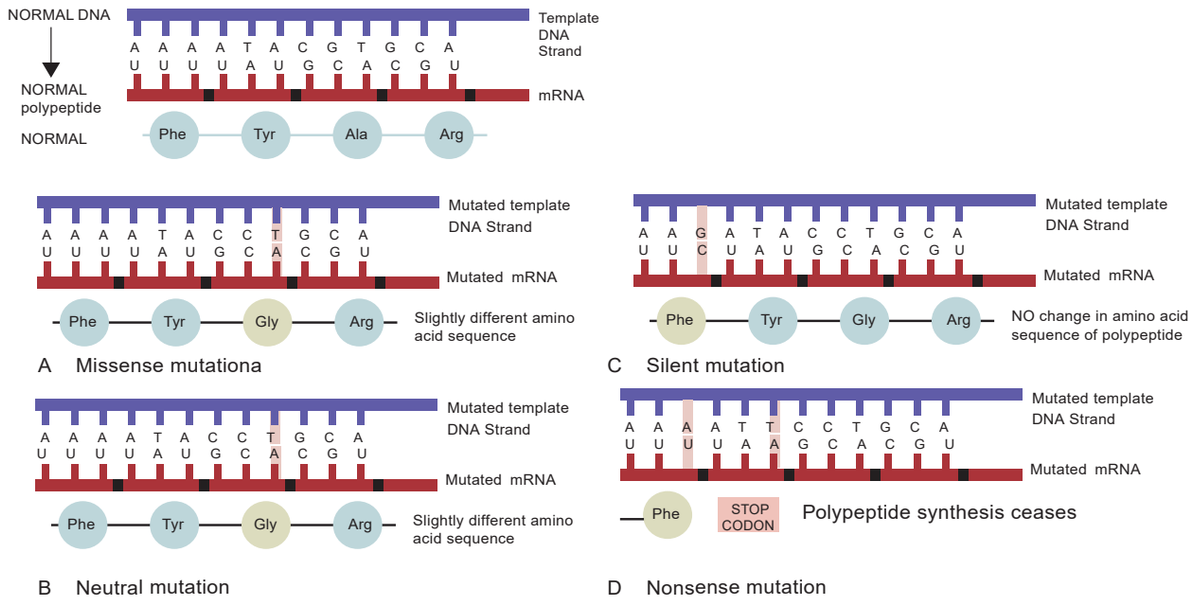


Figure 12.9: **(a)** Missense **(b)** neutral c) silent **(d)** nonsense mutation respectively

normal polypeptide fragments (often nonfunctional) are formed.

Forward mutations change the genotype from wild type to mutant and reverse mutations (or reversions or back mutations) change the genotype from mutant to wild type or to partially wild type. An organism which has reverted is Revertant. The effects of mutation may be diminished or abolished by a suppressor mutation. Suppressor mutation is a mutation at a different site from that of the original mutation. A suppressor mutation masks or compensates for the effects of the initial mutation, but it does not reverse the original mutation.

12.6 Formation of Mutants

The term mutant refers to an organism in which either the base sequence of DNA or the phenotype has been changed. A mutant is an organism whose genotype differs from that found in nature. The process of formation of mutant organism is called mutagenesis. In nature and in the laboratory, mutations sometimes arise spontaneously, without any help from the experimenter. This is called spontaneous mutagenesis. The two mechanisms that are most important for spontaneous mutagenesis are

1. Errors occurring during replication and
2. Spontaneous alteration of bases.

Mutations can also be induced experimentally by application of mutagens. Mutagens are agents that cause mutations.

Mutagens and their Mode of Action

Physical Mutagens

UV radiation: UV light causes mutations because the purine and pyrimidine bases in DNA absorb light strongly in the ultraviolet range (254 to 260 nm). At this wavelength, UV light induces point mutations primarily by causing photochemical changes in the DNA. One of the effects of UV radiation on DNA

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